Revision of the Grade 7 course

Angles on parallel lines, the angle sum of a triangle, isosceles triangles and the congruence tests — the facts every proof this quarter will quote.

LESSON 1–2 | 2 × 40 MIN | GEOMETRY 8, P. 3 | STAGE 9 · 5.1

LESSON CLOCK

5' | 14' | 7' | 12' | 2'

Warm-up	5 min
Explanation	14 min
Interactive	7 min
Practice	12 min
Homework	2 min

LEARNING OBJECTIVES

- Use vertically opposite, corresponding, alternate and co-interior angles.
- Use the angle sum of a triangle and the exterior angle theorem.
- State and use the properties of an isosceles triangle.
- Quote the four congruence tests correctly (SSS, SAS, ASA, RHS).

TEXTBOOK

National	pp. 3–4
Cambridge	Learner's Book pp. 104–108
Workbook	Workbook 5.1

01 Explanation

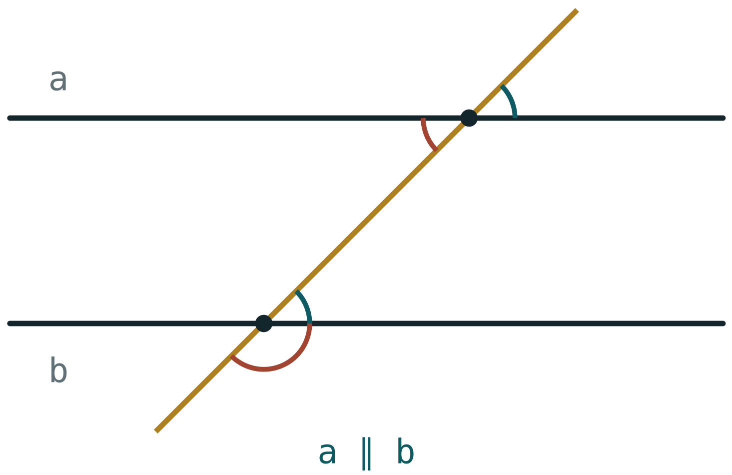
Why these four facts

Every proof in Chapter I — that the opposite sides of a parallelogram are equal, that the diagonals of a rectangle are equal, that the midline is half the base — is built from the same small set of Grade 7 facts. Two lessons spent making them automatic pays for the whole quarter.

1 · Angles on parallel lines

When a transversal crosses two parallel lines:

- **Corresponding** angles are equal (the “F” shape).
- **Alternate** angles are equal (the “Z” shape).
- **Co-interior** angles add to $180°$ (the “C” shape).



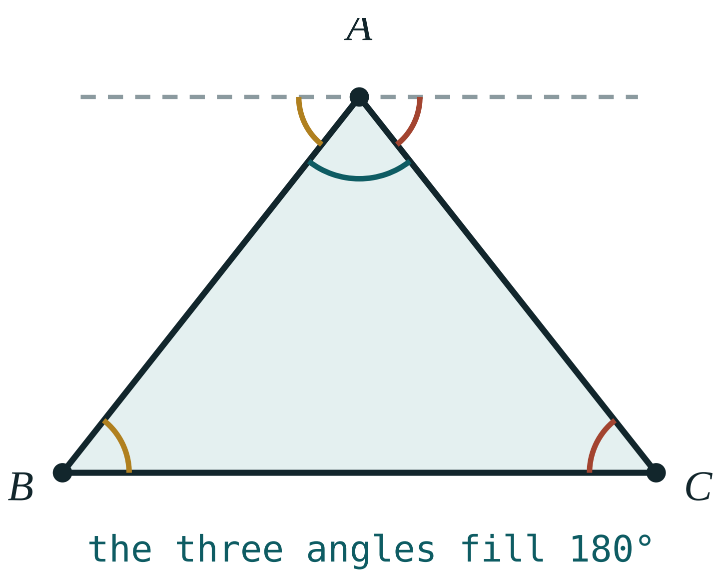
A transversal crossing two parallel lines. Equal angles are marked with the same colour.

And independently of any parallel lines: **vertically opposite** angles are equal, and angles on a straight line add to $180°$.

2 · The angle sum of a triangle

THEOREM

The three angles of any triangle add up to $180°$.



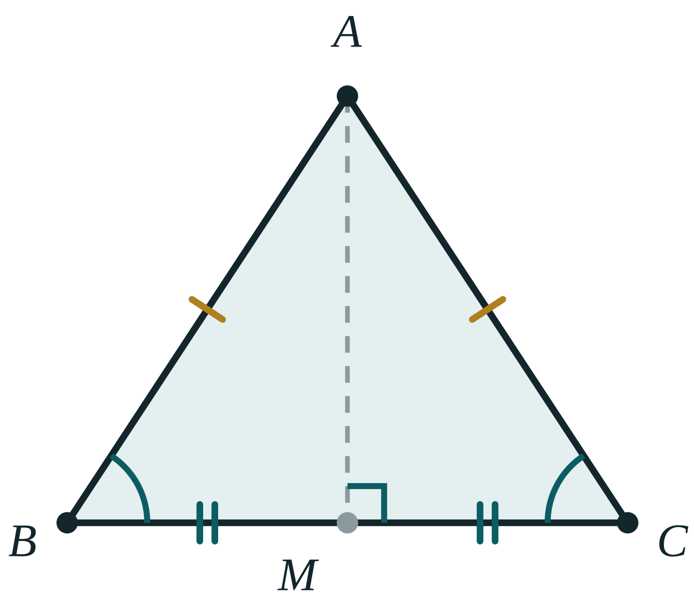
Draw a line through A parallel to BC. The two outer angles equal ∠B and ∠C (alternate angles), and together with ∠A they fill a straight line.

That figure *is* the proof: the dashed line through A is parallel to BC, so the two outer angles equal $\angle B$ and $\angle C$, and the three angles together make a straight line.

COROLLARY — THE EXTERIOR ANGLE

An exterior angle of a triangle equals the sum of the two interior angles not next to it.

3 · The isosceles triangle

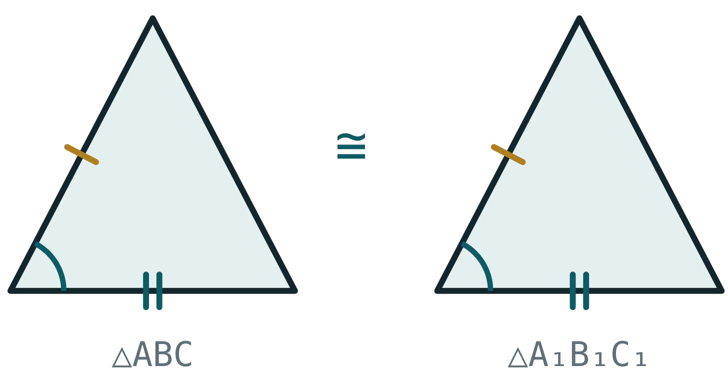


$$AB = AC \implies \angle B = \angle C$$

In an isosceles triangle the base angles are equal, and the line from the apex to the midpoint of the base is at the same time the median, the height and the angle bisector.

- $AB = AC \implies \angle B = \angle C$ — and the converse is also true.
- The median from the apex is also the height and the bisector of the apex angle.
- An equilateral triangle has all three angles $60°$.

4 · The congruence tests



Two triangles with two equal sides and the equal angle between them — the SAS test.

1. **SSS** — three sides.
2. **SAS** — two sides and the angle *between* them.
3. **ASA** — two angles and the side between them.
4. **RHS** — right angle, hypotenuse and one other side.

> THERE IS NO “SSA”
>
> Two sides and an angle that is *not* between them does not fix the triangle. Learners quote this constantly; correct it now rather than in the control work.

In congruent triangles, corresponding sides and corresponding angles are equal. Naming the vertices in the right order — $\triangle ABC \cong \triangle A_1B_1C_1$ — is what tells the reader which parts correspond.

02 Worked examples

EXAMPLE 1 ***In $\triangle ABC$, $\angle A = 52°$ and $\angle B = 61°$. Find $\angle C$ and the exterior angle at C.***

1. $\angle A + \angle B + \angle C = 180°$

   Angle sum of a triangle.

2. $\angle C = 180° - 52° - 61° = 67°$

   Subtract.

3. $exterior\ angle = 180° - 67° = 113°$

   Angles on a straight line.

4. $check\!: 52° + 61° = 113°$ ✓

   The exterior angle equals the two opposite interior angles.

> ANSWER
>
> $\angle C = 67°$, exterior angle $113°$

EXAMPLE 2 ***$\triangle ABC$ is isosceles with $AB = AC$ and $\angle A = 40°$. Find $\angle B$.***

1. $\angle B = \angle C$

   Base angles of an isosceles triangle.

2. $40° + 2\angle B = 180°$

   Angle sum, with $\angle C$ replaced by $\angle B$.

3. $2\angle B = 140°$

4. $\angle B = 70°$

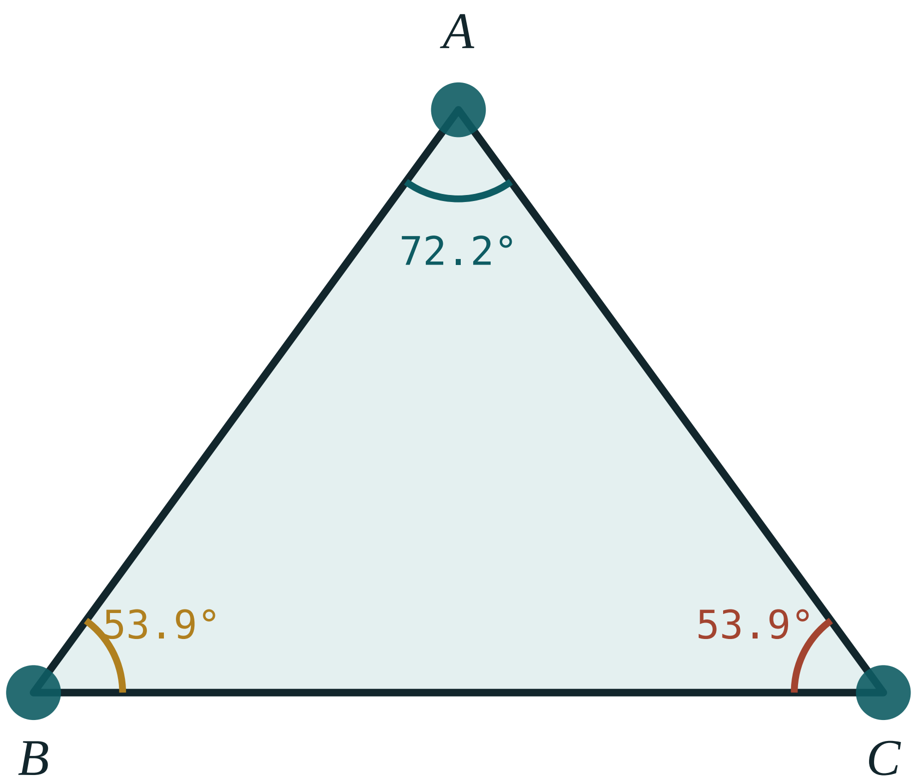
ANSWER

$\angle B = \angle C = 70°$

03 Interactive model

Drag a vertex until the triangle is nearly flat and ask what happens to the sum. It stays 180°.

The angles of a triangle



∠A **72.2°** | ∠B **53.9°** | ∠C **53.9°** | sum **180°** | longest side **BC** | largest angle **∠A**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Angle	Burchak	Угол
Vertex	Uchi	Вершина
Parallel lines	Parallel to‘g‘ri chiziqlar	Параллельные прямые
Transversal	Kesuvchi	Секущая
Corresponding angles	Mos burchaklar	Соответственные углы
Alternate angles	Almashinuvchi burchaklar	Накрест лежащие углы
Co-interior angles	Ichki bir tomonli burchaklar	Односторонние углы
Isosceles triangle	Teng yonli uchburchak	Равнобедренный треугольник
Congruent	Teng (mos)	Равные
Median	Mediana	Медиана
Bisector	Bissektrisa	Биссектриса
Altitude (height)	Balandlik	Высота

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Alternate angles on parallel lines are:

- equal
- supplementary
- complementary
- unrelated

Co-interior angles on parallel lines add to:

- $90°$
- $180°$
- $360°$
- they are equal

An exterior angle of a triangle equals:

- the nearest interior angle
- the sum of the two opposite interior angles
- $180°$
- half the angle sum

Which is NOT a congruence test?

- SSS
- SAS
- SSA
- ASA

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Two angles of a triangle are 40° and 75°. Find the third.*

   ANSWER *65°*

2. *An isosceles triangle has apex angle 40°. Find each base angle.*

   ANSWER *70°*

3. *Two parallel lines are cut by a transversal. One angle is 65°. Find its corresponding angle.*

   ANSWER *65°*

4. *Find the co-interior partner of an angle of 110°.*

   ANSWER *70°*

5. *Find the angle vertically opposite an angle of 128°.*

   ANSWER *128°*

6. *Each angle of an equilateral triangle is:*

   ANSWER *60°*

7. *An exterior angle of a triangle is 120°. Find the adjacent interior angle.*

   ANSWER *60°*

Medium · the standard question

7 PROBLEMS

1. *In $\triangle ABC$, $\angle A = 52°$ and $\angle B = 61°$. Find the exterior angle at C.*

   ANSWER $113°$

2. *An isosceles triangle has a base angle of 35°. Find the apex angle.*

   ANSWER $110°$

3. *The angles of a triangle are in the ratio 2 : 3 : 4. Find them.*

   ANSWER $40°, 60°, 80°$

4. *In $\triangle ABC$, $AB = AC$ and $\angle A = 2\angle B$. Find all three angles.*

   ANSWER $\angle A = 90°, \angle B = \angle C = 45°$

5. *Two angles of a triangle are equal and the third is 96°. Find them.*

   ANSWER $42°$ each

6. *Name the congruence test: two triangles share two sides and the angle between them.*

   ANSWER SAS

7. *Name the congruence test: two right triangles have equal hypotenuse and one equal leg.*

   ANSWER RHS

Hard · stretch and reason

7 PROBLEMS

1. *In $\triangle ABC$ the exterior angle at A is $130°$ and $\angle B = 55°$. Find $\angle C$.*

   ANSWER $\angle C = 75°$ — the exterior angle equals $\angle B + \angle C$.

2. *$\triangle ABC$ is isosceles with $AB = AC$. The bisector of $\angle B$ meets AC at D and $\angle ABD = 25°$. Find $\angle A$.*

   ANSWER $\angle B = 50°$, so $\angle C = 50°$ and $\angle A = 80°$

3. *Prove that the median to the base of an isosceles triangle is also its height.*

   ANSWER The two half-triangles are congruent by SSS, so the two angles at the base of the median are equal and together make $180°$ — each is $90°$.

4. *In $\triangle ABC$, $\angle A = 90°$ and $\angle B = \angle C$. Find $\angle B$, and say what kind of triangle it is.*

   ANSWER $45°$; a right isosceles triangle.

5. *Two parallel lines are cut by a transversal so that one pair of co-interior angles is in the ratio $4 : 5$. Find both.*

   ANSWER $80°$ and $100°$

6. *Explain why two triangles with equal angles need not be congruent.*

   ANSWER Equal angles fix the *shape* only. The triangles are similar; without one equal side they can be any size.

7. *$\triangle ABC$ has $\angle A = \angle B$. Prove that $CA = CB$.*

   ANSWER Converse of the isosceles-triangle theorem: drop the bisector of $\angle C$; the two triangles are congruent by ASA, so $CA = CB$.

07 Homework

Homework — 5 problems

Geometry 8, pp. 3–4 (revision). Draw a figure for every question.

1. *Two angles of a triangle are $38°$ and $47°$. Find the third and the exterior angle at that vertex.*
2. *An isosceles triangle has apex angle $96°$. Find the base angles.*

3. *The angles of a triangle are in the ratio 1 : 2 : 3. Find them and name the triangle.*
4. *A transversal cuts two parallel lines. One co-interior angle is 3 times the other. Find both.*
5. Write out the four congruence tests, each with a sketch.